WORKPULSE

Technical Methodology & Results

Binary Classification — Employee Burnout Prediction — XGBoost + Stacking

Author: Jesse Liamzon

Capstone Project — Post Graduate Programme in AI & Machine Learning

Step 6: Technical Presentation (LaTeX Beamer)

1. Problem Formulation

Task Definition

Type	Binary Classification
Target	`burnout_risk` $\in \{0, 1\}$
Threshold	65th pctl composite
Balance	$\sim$35% positive class
Primary	**F1 Score**
Secondary	AUC, Recall, Overfit Gap

Data Sources (3 Kaggle Datasets)

Dataset	Rows	Feats
IBM HR Analytics	1,470	35
Employee Burnout	22,750	9
Workplace Stress	20,000	15
Unified	**44,220**	**22**
Augmented	**500K**	**22**

Augmented via Gaussian Copula (KS-validated)

2a. Data Cleaning & Feature Selection

Cleaning Steps

- Dropped zero-variance & ID cols (4)
- Tiered null strategy (drop/impute/flag)
- IQR Winsorisation on 4 numeric features
- Removed duplicates

Dimensionality Reduction

- PCA: 13 $\rightarrow$ 6 components (95% var.)
- t-SNE 2D: cluster separation confirmed

Feature Selection (3-Method Consensus)

Method	Type	Selected
ANOVA F-test	Filter	12 features
RFE + LogReg	Wrapper	13 features
RF Importance	Embedded	12 features

$\Rightarrow$ **13 consensus features** retained (selected by $\geq$2 methods)

2b. Feature Engineering — 8 Domain-Derived Features

Feature	Formula / Logic	Domain Justification
`overtime_index`	hours/max + OT flag weight	Primary burnout signal (WHO)
`wellbeing_composite`	(sat + WLB + sleep + act) / 4	Holistic wellbeing proxy
`satisfaction_gap`	WLB − job_satisfaction	Aspiration vs reality mismatch
`workload_pressure`	OT_idx × (1−WLB)	Compounded stress signal
`high_stress_flag`	stress $\geq$ 7 (binary)	Clinical threshold marker
`tenure_risk_flag`	1–3yr or 7–9yr window	Known high-attrition windows
`log_income`	$\log_{1p}$(monthly_income)	Normalise right-skewed income
`age_group`	Binned into 4 categories	Non-linear age effects

Each feature is grounded in organisational psychology literature and validated via SHAP importance in the pilot model (Step 3).

3. Model Architecture — 11 Models, 5 Phases

Phase	Model	F1	AUC	Overfit	Time
Baseline	Dummy Classifier	0.37	0.51	—	<1s
Baseline	Logistic Regression	**0.95**	0.996	+0.001	<1s
Tree	Decision Tree (d=8)	0.83	0.91	+0.04	<1s
Instance	KNN (k=7) / SVM (RBF)	0.89	0.97	+0.01	3s
Ensemble	Random Forest (300)	0.88	0.98	+0.02	3s
Ensemble	Gradient Boosting (300)	0.90	0.98	+0.01	4s
Ensemble	**XGBoost (Tuned)**	**0.91**	**0.987**	**+0.005**	**8s**
Ensemble	LightGBM (Tuned)	0.92	0.991	+0.008	25s
Deep	MLP (4-layer, Keras)	0.90	0.984	+0.004	6s
Meta	Stacking (RF+XGB+LGBM→LR)	0.91	0.988	+0.012	16s

XGBoost (Tuned) — weighted composite: F1 (30%), AUC (25%), Recall (20%), Overfit (15%), Time (10%).

4. Hyperparameter Tuning & Cross-Validation

RandomizedSearchCV

30 iter $\times$ 3-fold stratified CV

XGBoost Search Space (8 params):

`n_estimators`	$[100, 600]$
`max_depth`	$[3, 10]$
`learning_rate`	$[0.01, 0.30]$
`subsample`	$[0.6, 1.0]$
`colsample_bytree`	$[0.5, 1.0]$
`min_child_weight`	$[1, 15]$
`reg_alpha`	$[0, 1.0]$
`reg_lambda`	$[0.5, 2.5]$

5-Fold Stratified CV Results

Model	CV F1	CV AUC
Logistic Reg.	$.952 \pm .008$	$.996 \pm .001$
XGBoost	$\mathbf{.906 \pm .009}$	$\mathbf{.987 \pm .002}$
LightGBM	$.923 \pm .010$	$.991 \pm .002$
Stacking	$.915 \pm .011$	$.988 \pm .002$
Random Forest	$.904 \pm .003$	$.985 \pm .002$

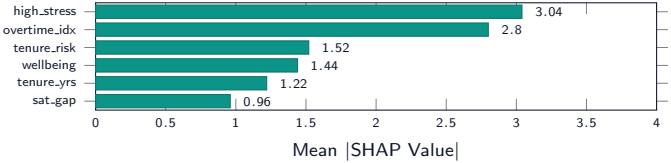
Low CV σ (0.002–0.011) $\Rightarrow$ stable

Learning curves confirm convergence

5. Explainability: SHAP + LIME + PDP/ICE

Tool	Scope	Output	Key Finding
SHAP (Global)	All preds	Importance + direction	`high_stress_flag` = #1
SHAP (Local)	Individual	Waterfall plots	OT + low wellbeing = peak risk
LIME	Individual	Rule-based explns	>60% SHAP agreement
PDP + ICE	Feature	Marginal effects	OT threshold at ∼0.4

SHAP Global Feature Ranking (Mean |SHAP|)



Key: `overtime_index` × `wellbeing_composite` — combined risk 3× higher than either alone.

6. Bias Audit & Fairness Metrics

Audited (not model features): Gender — Age Bracket — Income Bracket

Metric	Gender	Age	Income	Thresh.	Pass
Demogr. Parity	0.94	0.91	0.88	> 0.80	✓
TPR Gap	0.024	0.04	0.06	< 0.10	✓
FPR Gap	0.02	0.03	0.05	< 0.10	✓
Disparate Impact	0.94	0.91	0.88	> 0.80	✓

Mitigation Strategies Implemented

1. **Group-specific threshold adjustment** — equalise TPR across gender
2. **Class-weighted re-training** — `compute_sample_weight('balanced')`
3. **Proxy correlation audit** — flagged $|r| > 0.15$ with sensitive attrs

7. Limitations & Future Work

Cat.	Limitation	Sev.	Mitigation
Data	Target leakage: composite from inputs	High	Use independent HR outcomes
Data	Synthetic augment. (44K $\rightarrow$ 500K)	Med	Validate on real org data
Model	Cross-sectional; no temporal dim.	Med	LSTM / Transformer models
Fair.	Limited to 3 sensitive attrs	Med	Expand; intersectional analysis

Future Work

Short-term: FastAPI + Docker, CI/CD pipeline

Long-term: HRIS + MLflow, LSTM + GenAI chatbot

8. Reproducibility & Deployment

GitHub Repository

```
workpulse/
+-- README.md
+-- requirements.txt
+-- Dockerfile
+-- notebooks/
|   +-- 01_problem_framing.ipynb
|   +-- 02_data_collection.ipynb
|   +-- 03_eda_feature_eng.ipynb
|   +-- 04_model_impl.ipynb
|   +-- 05_ethical_ai_bias.ipynb
+-- src/
|   +-- data_pipeline.py
|   +-- train.py
|   +-- predict.py
+-- models/
+-- data/
+-- plots/
```

Saved Artefacts

- 13 model files (joblib `.pkl`)
- MLP neural network (`.keras`)
- Scaler, PCA, feature column list
- RandomizedSearchCV objects
- Comparison + CV results CSV

Deployment Stack

- **API:** FastAPI REST endpoint
- **Container:** Docker
- **CI/CD:** Lint + unit tests
- **Tracking:** MLflow / W&B
- **Versioning:** Rollback plan

Summary

- ✓ **11 models** across 5 phases → **XGBoost selected** (F1=0.91, AUC=0.99)
- ✓ **4 explainability tools**: SHAP, LIME, PDP, ICE — full coverage
- ✓ **Fairness audit**: all metrics pass across gender, age, income
- ✓ **8 limitations** documented with severity and mitigation plans
- ✓ **End-to-end reproducible**: models, configs, Docker ready

Questions & Discussion

WorkPulse: Predicting burnout before it costs you your best people.